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Additively manufactured footwear soles

Abstract

Articles of footwear comprising an additively manufactured sole. In some embodiments, the article of footwear may comprise a sole, an upper, and a joining element coupling the sole to the upper. In some embodiments, the sole may comprise a three-dimensional mesh with a ground-facing portion and a ledge formed on an upper side of the mesh. The ledge may comprise a continuous side surface that is coupled to the joining element. In some embodiments, the sole may comprise an undulating lower side comprising a plurality of bumps and a plurality of valleys positioned between respective bumps. In some embodiments, the sole may comprise a continuous ground-facing rim formed on a perimeter portion of a lower side of the sole.

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Background/Summary

FIELD

(1) Embodiments described herein generally relate to soles and sole features for an article of footwear. Specifically, embodiments described herein relate to additively manufactured soles with features designed to couple with an upper and/or provide ground contacting features.

BACKGROUND

(2) Footwear generally includes a sole that provides support and cushioning to a wearer's foot and an upper attached to the sole that encloses the wearer's foot. The sole may be constructed to provide the desired comfort and performance characteristics for the wearer. Soles may be made by molding a foam material, such as ethylene-vinyl acetate (EVA), among others.

(3) Runners and other athletes may desire footwear having specific performance characteristics to optimize their performance. Further, customization of the sole may allow the footwear to be tailored to a particular athlete. Thus, a continuing need exists for soles that provide desired properties and performance characteristics.

BRIEF SUMMARY

(4) Articles of footwear, and components thereof, according to the present disclosure may comprise one or more of the following features and combinations thereof.

(5) A first embodiment (1) of the present application is directed to an article of footwear comprising a sole comprising: a three-dimensional mesh extending from a toe region to a heel region of the sole, the mesh comprising: a lattice structure comprising a plurality of interconnected unit cells, each interconnected unit cell comprising a plurality of struts defining a three-dimensional structure and a plurality of nodes at which one or more unit cell struts are connected, a ground-facing portion comprising a plurality of traction elements and located at a lower side of the mesh opposite an upper side, and a ledge formed on the upper side of the mesh and comprising a continuous side surface extending toward the lower side and along at least one of an outer lateral side or an outer medial side of the mesh; an upper coupled to the mesh; and a joining element extending between and coupling the ledge to the upper.

(6) In a second embodiment (2), the mesh, the ledge, and the ground-facing portion according to

the first embodiment (1), are integrally formed as a single piece.

(7) In a third embodiment (3), the upper side of the mesh according to the first embodiment (1) or the second embodiment (2) comprises an upper surface with a substantially smooth overall surface contour.

(8) In a fourth embodiment (4), the continuous side surface of the ledge according to any one of embodiments (1)-(3) extends around the heel region of the sole from the outer lateral side to the outer medial side of the mesh.

(9) In a fifth embodiment (5), the ledge according to any one of embodiments (1)-(4) further comprises a solid top surface formed on an upper surface of the upper side of the mesh.

(10) In a sixth embodiment (6), the joining element according to any one of embodiments (1)-(5) is coupled to the continuous side surface of the ledge.

(11) In a seventh embodiment (7), the ledge according to any one of embodiments (1)-(6) extends into a midfoot region of the mesh.

(12) In an eighth embodiment (8), the lower side of the mesh according to any one of embodiments (1)-(7) comprises a plurality of bumps and a plurality of valleys positioned between respective bumps, and the plurality of traction elements of the ground-facing portion are formed on each of the plurality of bumps.

(13) In a ninth embodiment (9), each of the plurality of traction elements according to the eighth embodiment (8) comprises a separate continuous skin that covers a portion a respective one of the plurality of bumps.

(14) In a tenth embodiment (10), the lower side of the mesh according to any one of embodiments (1)-(7) comprises a ground-facing perimeter rim and a middle portion within the ground-facing perimeter rim, and the plurality of traction elements of the ground-facing portion are formed on the ground-facing perimeter rim.

(15) In an eleventh embodiment (11), the middle portion according to the tenth embodiment (10) is recessed relative to the perimeter rim.

(16) In a twelfth embodiment (12), the ground-facing perimeter rim according to the tenth embodiment (10) comprises a continuous surface skin that extends around each of the toe region, a midfoot region, and the heel region of the sole and the plurality of traction elements are formed on the continuous surface skin.

(17) A thirteenth embodiment (13) of the present application is directed to a sole for an article of footwear, the sole comprising: a three-dimensional mesh extending from a toe region to a heel region of the sole, the mesh comprising: a lattice structure comprising a plurality of interconnected unit cells, each interconnected unit cell comprising a plurality of struts defining a three-dimensional structure and a plurality of nodes at which one or more unit cell struts are connected, and an undulating lower side comprising a plurality of bumps and a plurality of valleys positioned between respective bumps, wherein the bumps are at least partially defined by one or more unit cells of the lattice structure; and a plurality of traction elements formed on the plurality of bumps of the undulating lower side of the mesh such that the plurality of traction elements are ground-facing.

(18) In a fourteenth embodiment (14), the plurality of bumps according to the thirteenth embodiment (13) comprise a first bump of the plurality of bumps located in the toe region and comprising a first height measured between a first peak of the first bump and a bottom of an adjacent valley of the plurality of valleys, and a second bump of the plurality of bumps located in the heel region and comprising a second height measured between a second peak of the second bump and a bottom of an adjacent valley of the plurality of valleys, wherein the second height is different from the first height.

(19) In a fifteenth embodiment (15), the second height according to the fourteenth embodiment (14) is larger than the first height.

(20) In a sixteenth embodiment (16), each one of the plurality of traction elements according any one of embodiments (13)-(15) comprises a separate continuous skin that covers a portion a

respective one of the plurality of bumps.

(21) In a seventeenth embodiment (17), the continuous skin of each traction element according to the sixteenth embodiment (16) comprises a ribbed traction pattern.

(22) In an eighteenth embodiment (18), at least one of the bumps of the plurality of bumps located in a toe region of the sole according to any one of embodiments (13)-(17) comprises a flatter contour than at least one of the bumps of the plurality of bumps located in the heel region of the sole.

(23) In a nineteenth embodiment (19), the sole according to any of one embodiments (13)-(18) further comprises a plurality of connecting ribs formed on the undulating lower side of the mesh and extending between adjacent traction elements of the plurality of traction elements.

(24) A twentieth embodiment (20) of the present application is directed to an sole for an article of footwear, the sole comprising: a three-dimensional mesh extending from a toe region to a heel region of the sole, the mesh comprising: a lattice structure comprising a plurality of interconnected unit cells, each interconnected unit cell comprising a plurality of struts defining a three-dimensional structure and a plurality of nodes at which one or more unit cell struts are connected; and a continuous ground-facing rim formed on a perimeter portion of a lower side of the mesh, wherein the continuous ground-facing rim extends around each of the toe region, a midfoot region, and the heel region of the sole, and wherein the continuous ground-facing rim comprises a plurality of traction elements formed on a ground-facing surface of the rim.

(25) In a twenty-first embodiment (21), a recess according to the twentieth embodiment (20) is formed in the midfoot region of a middle portion of the mesh between the ground-facing rim.

(26) In a twenty-second embodiment (22), the plurality of traction elements according to the twentieth embodiment (20) or the twenty-first embodiment (21) form a ribbed traction pattern.

(27) In a twenty-third embodiment (23), the sole according to any one of embodiments (20)-(22) further comprises a plurality of connecting ribs formed on the mesh and extending through a middle portion of the lower side to connect a lateral side of the ground-facing rim to a medial side of the ground-facing rim.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles thereof and to enable a person skilled in the pertinent art to make and use the same.

(2) FIG. 1 illustrates a side view of an article of footwear having a sole coupled to an upper with a joining element, according to some embodiments.

(3) FIG. 2 illustrates an exploded view of an article of footwear including a sole, an upper, and a joining element, according to some embodiments.

(4) FIG. 3 illustrates a cross-sectional diagram of an assembled article of footwear showing a sole coupled to an upper with a joining element, according to some embodiments.

(5) FIG. 4A illustrates a side view of a sole, according to some embodiments.

(6) FIG. 4B illustrates a top-down view of the sole shown in FIG. 4A.

(7) FIG. 5 illustrates a perspective top view of a sole, according to some embodiments.

(8) FIG. 6 illustrates a perspective bottom view of the sole shown in FIG. 5.

(9) FIG. 7 illustrates a bottom-up view of the sole shown in FIG. 5.

(10) FIG. 8 illustrates a cross-sectional view of the sole in FIG. 7 along cross-sectional line 8-8.

(11) FIG. 9 illustrates a cross-sectional view of the sole in FIG. 7 along cross-sectional line 9-9.

(12) FIG. 10 illustrates a perspective bottom view of a sole, according to some embodiments.

(13) FIG. 11 illustrates a bottom-up view of the sole shown in FIG. 10.

(14) FIG. 12 illustrates a cross-sectional view of the sole in FIG. 11 along cross-sectional line 12-12.

DETAILED DESCRIPTION

(15) Reference will now be made in detail to representative embodiments illustrated in the accompanying drawing. It should be understood that the following descriptions are not intended to limit the embodiments to one preferred embodiment. To the contrary, it is intended to cover alternatives, modifications, and equivalents as can be included within the spirit and scope of the described embodiments as defined by the claims.

(16) References in the specification to “some embodiments,” “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

(17) The indefinite articles “a,” “an,” and “the” include plural referents unless clearly contradicted or the context clearly dictates otherwise.

(18) The term “comprising” is an open-ended transitional phrase. A list of elements following the transitional phrase “comprising” is a non-exclusive list, such that elements in addition to those specifically recited in the list can also be present.

(19) As used herein, unless specified otherwise, references to “first,” “second,” “third,” “fourth,” etc. are not intended to denote order, or that an earlier-numbered feature is required for a later-numbered feature. Also, unless specified otherwise, the use of “first,” “second,” “third,” “fourth,” etc. does not necessarily mean that the “first,” “second,” “third,” “fourth,” etc. features have different properties or values.

(20) Soles and midsoles of footwear can be formed by molding methods, such as by injection or compression molding. In some cases, when midsoles are molded in one piece, the properties of the resulting midsole cannot be made to vary across different portions of the midsole. As a result, the molded midsole may comprise isotropic properties. However, in some cases, it may be desirable to provide a midsole with mechanical properties that vary across or within different regions and/or that vary depending on the directions in which the midsole is loaded to improve the performance of the midsole and allow for customization of the performance of the midsole. For example, it may be desirable to provide a midsole with anisotropic properties that vary on different portions of the midsole in order to improve the performance of the midsole and allow for customization of the performance of the midsole.

(21) Some embodiments described herein relate to footwear comprising a sole that comprises a three-dimensional mesh to provide the sole with desired properties, for example ground-contacting properties and anisotropic properties. In some embodiments, the mesh may be customized to provide different properties in different regions of the sole. In some embodiments, anisotropic properties may help to guide a foot of an athlete during sports movements, or may be used to guide a foot of a wearer in daily use. Further, selective mechanical deformation of the mesh may be achieved to provide stride length gains during phases of ground contact while walking or running. Such stride length gains can be optimized by selection of the geometry and dimensions of the mesh. In some embodiments, the mesh may absorb midfoot and heel strike forces and translate vertical momentum in running into forward momentum through angular-biased mesh features arranged to translate force applied in a desired direction and create angular rotation.

(22) Some embodiments described herein relate to an article of footwear or a footwear component that comprises a sole comprising an undulating or irregular bottom surface formed on a lower side

of a mesh to provide a plurality of traction elements dispersed across the lower side of the sole. As a result, footwear can be customized to provide the sole with mechanical properties that vary across or within different regions and/or that vary depending on the direction in which the midsole is loaded (for example, anisotropic properties) to provide performance improvements. Some embodiments described herein relate to an article of footwear or a footwear component comprising a three-dimensional mesh that is additively manufactured. The additive manufacturing techniques described herein can create footwear comprising custom properties produced by controlling the mesh geometry and dimensions.

(23) Some embodiments described herein relate to an article of footwear or a footwear component comprising a sole comprising a substantially smooth or flat top surface. During manufacture of the sole, an upper side of the sole may be coupled to a build plate or printing head for additively manufacturing sole. Such a manufacturing configuration allows for additional freedom of design and construction of the lower side of the sole. For example, as described herein, the lower side (ground-facing side) can be customized to meet a user's desire for support or performance in the sole and/or can comprise an undulating or irregular bottom surface as described herein. Further, the lower side (ground-facing side) can be designed with peaks, valleys, and/or recesses as described herein. These peaks, valleys, and/or recesses can provide, among other things, cushioning properties, performance properties, and/or weight saving.

(24) In addition to a customized bottom surface, the smooth top surface can also provide benefits in the assembly of the article of footwear. In some embodiments, the flat top surface may comprise integrally formed solid surfaces that increase a contact area for adhesion with other components of the article of footwear. In some embodiments, a ledge may be formed in the smooth or flat top surface that may comprise a large continuous surface area to adhere with a joining element, which in turn may couple with the upper. This large continuous surface may increase a bond strength of adhesion between the joining element and the ledge of the sole, compared to a bond strength of adhesion between the joining element and the mesh pattern in the sole.

(25) As used herein, the term three-dimensional mesh refers to a three-dimensional structure comprising a plurality of interconnected unit cells arranged in a web-like structure or a lattice structure. The web-like or lattice structure of a mesh comprises interconnected structural members (struts) defining the plurality of unit cells. The structural members, and thus the unit cells, may be connected at nodes. For example, the interconnected structural members may be struts that are connected at nodes and that define unit cells arranged in a lattice configuration. In some embodiments, the plurality of interconnected unit cells may be arranged in a regular or repeating lattice configuration. Exemplary lattice configurations include, but are not limited to, basic cubic lattices, body-centered cubic lattices, face-centered cubic lattices, and modified lattices based on these lattice types. Exemplary lattice configurations include, but are not limited to, the lattice structures described in U.S. patent application Ser. Nos. 17/069,623 and 18/313,135, which are hereby incorporated by reference in their entireties.

(26) Unit cells may comprise any of various dimensions and geometries. Further, unit cells within a three-dimensional mesh may be the same or may differ. Thus, a mesh may comprise unit cells of different dimensions or geometries. The three-dimensional shape of a unit cell may be defined by a plurality of interconnected struts connected to one another at nodes. In such embodiments, each unit cell may comprise a base geometry defined by the struts. As used herein, "base geometry" means the base three-dimensional shape, connection, and arrangement of the struts defining a unit cell. The base geometry of a unit cell may be, but is not limited to, a dodecahedron (for example, rhombic), a tetrahedron, an icosahedron, a cube, a cuboid, a prism, or a parallelepiped. Each node may connect two or more struts. Struts may be arranged to provide a mesh with the desired performance characteristics, and a mesh may comprise regions with different densities of struts.

(27) In some embodiments, the interconnected unit cells may comprise a solid representation of a repeating implicit surface of a lattice structure. In such embodiments, the unit cells may comprise a

“base surface geometry” defined by the base three-dimensional shape of a body formed by one or more ribbons (walls) of material that define a solid representation of an implicit surface for a full unit cell. In some embodiments, the implicit surface may be a periodic implicit surface such that the base surface geometry of each unit cell contacts the base surface geometry of at least some adjacent unit cells to create a lattice. One example of a suitable periodic surface is a gyroid, but any type of suitable periodic surface can be used.

(28) Herein, a solid representation of an implicit surface refers to a solid object following the shape of an implicit surface. Whereas an actual implicit surface has no thickness, a solid representation of an implicit surface has a thickness on one or both sides of the actual implicit surface in a three dimensional space. The thickness gives the solid representation volume, meaning the solid representation can be built as a physical object from physical material. The added thickness or thicknesses may be uniform, or at least approximately uniform notwithstanding fillets or local deformities, and thin in comparison to the overall size of the represented implicit surface. In some embodiments, the relative density of a unit cell of the solid representation may be from 5% to 30%, from 5% to 40%, from 10% to 25%, or from 15% to 20%. The term “relative density” as used herein refers to an amount of a unit cell occupied by solid material as a percentage of a total volume of the unit cell.

(29) In some embodiments, the implicit surfaces may be created using a combination of random Fourier series functions, in which linear and or nonlinear coefficient as well as linear and nonlinear variables inside sinuous and cosine terms over the x, y and z space are iterated to generate the functions. The resulting unit cells may have different planes of symmetry, such as, in various examples, zero planes of symmetry, one plane of symmetry, or more than one plane of symmetry. The function may be derived in a way that satisfies the periodicity of the unit cell. Criteria for the selection of an applicable implicit surface within the design space domain may comprise any one or any combination of number of terms in the equation, number of connected components, the edge boundary length, surface area, and volume fraction.

(30) Soles for articles of footwear described herein (for example, sole **112**), and any component of the soles footwear described herein may be formed by additive manufacturing (for example, three-dimensional (3D) printing). Exemplary additive manufacturing techniques include for example, selective laser sintering, selective laser melting, selective heat sintering, stereo lithography, fused deposition modeling, or 3-D printing in general. Various additive manufacturing techniques related to articles of footwear are described for example in US 2009/0126225, WO 2010/126708, US 2014/0300676, US 2014/0300675, US 2014/0299009, US 2014/0026773, US 2014/0029030, WO 2014/008331, WO 2014/015037, US 2014/0020191, EP 2564719, EP 2424398, and US 2012/0117825. In some embodiments, the additive manufacturing process may include a continuous liquid interface production process. For example, the additive manufacturing process may include a continuous liquid interface production process as described in U.S. Pat. No. 9,453,142, issued on Sep. 27, 2016, which is hereby incorporated in its entirety by reference thereto.

(31) In some embodiments, 3-D printing a sole for an article of footwear, or component thereof, may comprise 3-D printing the sole or component in an intermediate green state, shaping the sole or component in the green state, and curing the green state in its final shape. In some embodiments, 3-D printing a sole for an article of footwear, or component thereof, may comprise 3-D printing the sole or component in an intermediate green state, expanding the intermediate green state, shaping the sole or component in the green state, and curing the green state in its final shape.

(32) Techniques for producing an intermediate green state object from resins by additive manufacturing are known. Suitable techniques include bottom-up and top-down additive manufacturing, generally known as stereolithography. Such methods are known and described in, for example, U.S. Pat. No. 5,236,637 to Hull, U.S. Pat. Nos. 5,391,072 and 5,529,473 to Lawton, U.S. Pat. No. 7,438,846 to John, U.S. Pat. No. 7,892,474 to Shkolnik, U.S. Pat. No. 8,110,135 to

El-Siblani, U.S. Patent Application Publication No. 2013/0292862 to Joyce, and US Patent Application Publication No. 2013/0295212 to Chen et al. The disclosures of these patents and applications are incorporated by reference herein in their entirety.

(33) In some embodiments, the additive manufacturing step may be carried out by one of the family of methods sometimes referred to as continuous liquid interface production (CLIP). CLIP is known and described in, for example, U.S. Pat. Nos. 9,211,678; 9,205,601; 9,216,546; and others; in J. Tumbleston et al., Continuous liquid interface production of 3D Objects, *Science* 347, 1349-1352 (2015); and in R. Janusiewicz et al., Layerless fabrication with continuous liquid interface production, *Proc. Natl. Acad. Sci. USA* 113, 11703-11708 (Oct. 18, 2016). Other examples of methods and apparatus for carrying out particular embodiments of CLIP include, but are not limited to: Batchelder et al., US Patent Application Pub. No. US 2017/0129169 (May 11, 2017); Sun and Lichkus, US Patent Application Pub. No. US 2016/0288376 (Oct. 6, 2016); Willis et al., US Patent Application Pub. No. US 2015/0360419 (Dec. 17, 2015); Lin et al., US Patent Application Pub. No. US 2015/0331402 (Nov. 19, 2015); D. Castanon, US Patent Application Pub. No. US 2017/0129167 (May 11, 2017). B. Feller, US Pat App. Pub. No. US 2018/0243976 (published Aug. 30, 2018); M. Panzer and J. Tumbleston, US Pat App Pub. No. US 2018/0126630 (published May 10, 2018); K. Willis and B. Adzima, US Pat App Pub. No. US 2018/0290374 (Oct. 11, 2018) L. Robeson et al., PCT Patent Pub. No. WO 2015/164234 (see also U.S. Pat. Nos. 10,259,171 and 10,434,706); and C. Mirkin et al., PCT Patent Pub. No. WO 2017/210298 (see also US Pat. App. US 2019/0160733). The disclosures of these patents and applications are incorporated by reference herein in their entirety.

(34) While stereolithography techniques such as CLIP may be preferred, it will be appreciated that other additive manufacturing techniques, such as jet printing (see, e.g., U.S. Pat. No. 6,259,962 to Gothait and US Patent App. Ser. No. US 2020/0156308 to Ramos et al.) may also be used.

(35) In any of the embodiments described herein, a mesh may be selected to provide desired performance characteristics. A mesh may be tailored to provide a higher stiffness to weight ratio to provide a lightweight midsole, to control midsole shear stiffness to allow for or to prevent midsole shear, and to control energy return and damping.

(36) Exemplary materials for sole **112** and components thereof (for example, mesh **130**) include, but are not limited to, a foam, a rubber, ethyl vinyl acetate (EVA), a thermoplastic elastomer, a thermoplastic polyurethane (TPU), an expanded thermoplastic polyurethane (eTPU), an expanded elastomeric polyurethane, a polyether block amide (PEBA), an expanded polyether block amide (ePEBA), a thermoplastic rubber (TPR), and a polyolefin, for example polyethylene (PE), polystyrene (PS), polypropylene (PP), nylon (polyamides), carbon, graphene, carbon fiber, carbon nanotubes, fiber reinforced polymers, mycelium, aluminum, steel, titanium, or any other suitable material.

(37) Some embodiments described herein relate to an article of footwear comprising an upper **110**, a sole **112**, and a joining element **114** configured to couple the sole **112** to the upper **110**, as shown, for example, in FIG. **1**. In some embodiments, article of footwear **100** may be a footwear component, such as a sole or a midsole, or may be a complete article of footwear, such as a shoe, sneaker, boot, or cleat, among other types of footwear. In the illustrative embodiment shown in FIGS. **1** and **2**, joining element **114** overlaps with a portion of sole **112** and a portion of upper **110** to couple the upper **110** and sole **112** together. In some embodiments, joining element **114** can increase a bond strength between upper **110** and sole **112**. In some embodiments, sole **112** may comprise a three-dimensional mesh **130** such that adjacent surfaces to upper **110** in article of footwear **100** are formed by lattice structure **140**. In the illustrative embodiments shown in FIGS. **4A**, **4B**, and **5**, a ledge **134** is formed around a heel region **124** of sole **112**. Ledge **134** may provide a continuous surface for joining element **114** to adhere to.

(38) In some embodiments, upper **110** may comprise a non-woven fabric, a woven fabric, or knitted textile material. In some embodiments, upper **110** may comprise a knit upper. A knit upper may be

formed by flat knitting or circular knitting. In some embodiments, upper **110** may comprise a sock-type upper. Exemplary textile materials for upper **110** comprise, but are not limited to, thermoplastic polyurethane (TPU), polyester, polyamide, polyethylene (PE), PE foam, polyurethane (PU) foam, nylon, ultra-high molecular weight polyethylene (for example, DYNEMA® (a type of ultra-high molecular weight polyethylene)), carbon fiber, KEVLAR® (a type of para-aramid), synthetic spider silk, cotton, wool, natural or artificial silk, polyethersulfone (PES), ELASTAN® (a polyether-polyurea copolymer), or a blend of two or more of these materials.

(39) In some embodiments, sole **112** may comprise three-dimensional mesh **130**, a ground-facing portion **132**, and ledge **134**, as shown, for example, in FIGS. **4A**, **4B**, and **5**. In some embodiments, mesh **130**, ground-facing portion **132**, and ledge **134** may be integrally formed (for example integrally 3D printed) as a single piece sole **112**. In some embodiments, at least two of mesh **130**, ground-facing portion **132**, and ledge **134** may be assembled and bonded together to form sole **112**. Sole **112** may comprise a toe region **120**, a midfoot region **122**, a heel region **124**, a medial side **126**, and a lateral side **128**, as shown, for example in FIGS. **4A** and **4B**.

(40) Three-dimensional mesh **130** may provide sole **112** with desired properties, for example anisotropic properties, and/or allow sole **112** to be customized to provide different properties in different regions **120**, **122**, **124** of sole **112**. For example, selective mechanical deformation of mesh **130** may provide stride length gains during phases of ground contact while walking or running, which may be optimized by selection of the geometry and dimensions of mesh **130**. In addition, mesh **130** may be configured to absorb midfoot and heel strike forces and translate vertical momentum in running into forward momentum through angular-biased features arranged to translate force applied in a desired direction and create angular rotation.

(41) In some embodiments, mesh **130** may be formed by any of the additive manufacturing (for example, three-dimensional (3D) printing) techniques described herein.

(42) In the illustrative embodiment shown in FIGS. **1**, **2**, **4A**, **4B**, and **5**, mesh **130** may comprise a plurality of interconnected unit cells **140** arranged in a web-like structure or a lattice structure. The plurality of unit cells **140** may be formed by a plurality of struts **142** connected together at a plurality of nodes **144**. The plurality of struts **142** may be interconnected structural members that define a three-dimensional structure in mesh **130**, and may determine the desired properties of sole **112**. The plurality of struts **142** are connected at the plurality of nodes **144** and define unit cells **140** of mesh **130**. In some embodiments, the plurality of interconnected unit cells **140** may be arranged in a regular or repeating lattice configuration.

(43) In some embodiments, mesh **130** may comprise upper side **150** and lower side **152** as shown, for example, in FIGS. **4A-6**. Upper side **150** may comprise upper surface **154** configured to contact upper **110** when the article of footwear **100** is assembled, and an upper perimeter edge **155** extending around the perimeter of upper surface **154**. In some embodiments, upper surface **154** may comprise an upper perimeter rim **153** that forms a continuous surface around the perimeter edge **155** of the upper side **150**. In some embodiments, a portion of unit cells **140** may be exposed on upper side **150** between the edges of the upper perimeter rim **153**. In some embodiments, upper surface **154** may comprise a substantially smooth overall surface such the plurality of struts **142** have a flat profile at the upper surface **154**. In some embodiments, upper surface **154** may comprise a shallow contour that follows an approximate shape of a user's foot. In some embodiments, upper surface **154** may have a first radius in toe region **120** and a second radius in midfoot region **122**. In some embodiments, the first radius may be about 410 mm. In some embodiments, the first radius may be greater than or equal to about 400 mm to less than or equal to about 420 mm. In some embodiments, the second radius may be about 100 mm. In some embodiments, the second radius may be greater than or equal to about 90 mm to less than or equal to about 110 mm. In some embodiments, upper surface **154** may have a shape formed by a method as described in U.S. application Ser. No. 17/419,601, the disclosure of which is incorporated by reference herein in its

entirety.

(44) In some embodiments, upper perimeter rim **153** may comprise a toe bumper **151** internally formed with (for example, integrally 3D printed with) upper perimeter rim **153**.

(45) Ledge **134** may be formed on upper side **150** of sole **112** around heel region **124** as shown, for example, in FIGS. **4B** and **5**. Ledge **134** may be configured to provide a continuous surface **156**, **158** to adhere with joining element **114**. Ledge **134** may comprise a large continuous surface area to adhere with joining element **114**. The continuous surface area of ledge **134** may increase a bond strength of adhesion between joining element **114** and ledge **134**, compared to a bond strength of adhesion between joining element **114** and a plurality of interconnected unit cells **140** of mesh **130**.

(46) Ledge **134** may comprise a continuous side surface **156**. In some embodiments, ledge may comprise a solid top surface **158** as shown, for example, in FIGS. **4A**, **4B**, and **5**.

(47) Solid top surface **158** may extend around heel region **124** and along upper surface **154** away from upper perimeter edge **155**. In some embodiments, solid top surface **158** may be approximately perpendicular to continuous side surface **156**. In some embodiments, solid top surface **158** may extend away from continuous side surface **156** at an obtuse angle relative to continuous side surface **156**. In some embodiments, solid top surface **158** may form part of upper perimeter rim **153**. In some embodiments, solid top surface **158** may extend beyond upper perimeter rim **153** and further from upper perimeter edge **155** than upper perimeter rim **153**.

(48) Continuous side surface **156** may extend around heel region **124** and away from upper perimeter edge **155** to a lower ledge edge **189** of ledge **134** along the outer medial side **126**, the outer lateral side **128**, or both, as shown, for example, in FIG. **5**. In the illustrative embodiment shown in FIGS. **4B** and **5**, continuous side surface **156** extends along a portion of medial side **126** and lateral side **128** of sole **112**, extending from the heel region **124** and into a midfoot region **122** of sole **112**. In some embodiments, continuous side surface **156** may extend further along medial side **126** than lateral side **128**. In some embodiments, continuous side surface **156** may extend further along lateral side **128** than medial side **126**. In some embodiments, lower ledge edge **189** may be used to align, correctly position, and/or orient joining element **114** when sole **112** and upper **110** are coupled together.

(49) In some embodiments, continuous side surface **156** extends a ledge distance **159** from upper perimeter edge **155** to a lower ledge edge **189**, between upper side **150** and lower side **152**. In some embodiments, ledge distance **159** may be greater than or equal to about 3 millimeters (mm) to less than or equal to about 15 mm from upper perimeter edge **155**. In some embodiments, ledge distance **159** may be greater than or equal to about 5 mm to less than or equal to about 10 mm from upper perimeter edge **155**. In some embodiments, ledge distance **159** may be about 10 mm from upper perimeter edge **155**. In some embodiments, ledge distance **159** may be characterized as a portion of the distance between the upper side **150** and lower side **152** of sole **112**. In some embodiments, ledge distance **159** may be greater than or equal to about 10% to less than or equal to about 50% of the distance between the upper side **150** and lower side **152** of sole **112**. In some embodiments, ledge distance **159** may be greater than or equal to about 15% to less than or equal to about 40% of the distance between the upper side **150** and lower side **152** of sole **112**.

(50) In some embodiments, continuous side surface **156** is a solid component extending unbroken around sole **112**. In some embodiments, a plurality of apertures **157** may be formed in continuous side surface **156** along the medial and lateral sides **126**, **128** of sole **112**. In some embodiments, the plurality of apertures **157** may allow for venting during manufacture of sole **112** to improve yield of sole **112** and reduce defects. In some embodiments, the plurality of apertures **157** provides venting during manufacture of sole **112** that may reduce manufacturing lead times for sole **112**.

(51) In the illustrative embodiments shown in FIGS. **1-3**, upper **110** is coupled to sole **112** by joining element **114**. In some embodiments, joining element **114** may be coupled to sole **112** with an adhesive **116**, such as an epoxy, a glue, or any other suitable adhesives. Joining element **114** overlaps and is adhered to a portion of upper **110** with adhesive **116**, and overlaps and is adhered to

ledge **134** of sole **112** with adhesive **116**. Overlapping of joining element **114** across upper **110** and ledge **134** of sole **112** may help to facilitate securement of sole **112** to upper **110**. Ledge **134** may provide a greater adhesion contact area than unit cells **140** of mesh **130**, and thereby allow for increased bonding strength between joining element **114** and sole **112**. In some embodiments, joining element **114** may comprise foxing tape. In some embodiments, joining element **114** may comprise a flange that couples sole **112** to upper **110**. In some embodiments, joining element **114** may comprise a molded clip. Suitable materials for joining element **114** may include, but are not limited to, a rubber, ethyl vinyl acetate (EVA), a thermoplastic elastomer, a thermoplastic polyurethane (TPU), an expanded thermoplastic polyurethane (eTPU), an expanded elastomeric polyurethane, a polyether block amide (PEBA), an expanded polyether block amide (ePEBA), a thermoplastic rubber (TPR), and a polyolefin, for example polyethylene (PE), polystyrene (PS) or polypropylene (PP), nylon (polyamides), or a foam comprising one or more of these materials. Additional suitable materials include graphene, carbon fiber, carbon nanotubes, fiber reinforced polymers, mycelium, aluminum, steel, titanium, or any other suitable material. In some embodiments, joining element **114** may comprise a material partially or fully melted to couple upper **110** and sole **112** together, for example, by infrared welding.

(52) An illustrative example of the assembly of upper **110**, sole **112**, and joining element **114** is shown in FIG. 3. Upper **110** is assembled on upper surface **154** of mesh **130**. Upper **110** may comprise a shape in the portions adjacent to sole **112** that contour to a user's foot. This shape may form gaps **111** between upper **110** and sole **112**. In some embodiments, the gaps **111** are filled with adhesive **116** and covered by joining element **114** during construction of the article of footwear **100**. In some embodiments, joining element **114** may form an outermost surface of the article of footwear **100** extending between the overlapping portion of upper **110** and ledge **134**.

(53) In some embodiments, as illustrated in for example FIGS. 5-9, lower side **152** of mesh **130** may comprise an undulating surface such that ground-facing portion **132** formed on lower side **152** of mesh **130**. In some embodiments, lower side **152** of mesh **130** may comprise a plurality of bumps **160** and a plurality of valleys **170** positioned between respective bumps of the plurality of bumps **160**. The plurality of bumps **160** are at least partially formed by one or more underlying unit cells **140** of mesh **130**. Similarly, the plurality of valleys **170** are at least partially formed by one or more underlying unit cells **140** of mesh **130**.

(54) The plurality of bumps **160** may be dispersed across lower side **152** in a uniform or non-uniform pattern. In some embodiments, the plurality of bumps **160** may be dispersed across lower side **152** to provide support to desired portions of a user's foot. In some embodiments, the plurality of bumps **160** may be positioned to enhance performance of sole **112** for a user. In some embodiments, the underlying one or more unit cells **140** beneath the plurality of bumps **160** may comprise selective mechanical deformation for each respective bump **160** of the plurality of bumps **160**. For example, a bump **162**, **164** may absorb a strike force and translate vertical momentum in running into forward momentum through angular-biased mesh geometry arranged to translate force applied in a desired direction and create angular rotation.

(55) In some embodiments, ground-facing portion **132** may comprise a plurality of traction elements **190** formed on each of the plurality of bumps **160** as shown, for example, in FIGS. 6-9. In some embodiments, each of the plurality of traction elements **190** may form a continuous skin **192** that covers a portion of a respective one of the plurality of bumps **160**. In some embodiments, each continuous skin **192** may extend over each respective bump **160** and be configured to contact the ground during use. In such embodiments, sole **112** can be devoid of an outsole material disposed on skins **192**. In some embodiments, each continuous skin **192** may be integrally formed (for example, integrally 3D printed) with mesh **130**.

(56) In some embodiments, each continuous skin **192** may comprise a traction pattern **194**. In such embodiments, traction pattern **194** may serve to increase grip when sole **112** contacts the ground during use. In the illustrative embodiment in FIGS. 6 and 7, traction pattern **194** may comprise a

ribbed traction pattern comprising a plurality of contoured ribs. In some embodiments, traction pattern **194** may comprise concentric ribbed circles, overlapping ribbed circles, wave shaped ribs, or any other suitable geometric shape. In some embodiments, traction pattern **194** may correspond to the sport the article of footwear **100** is used for. For example, traction pattern **194** may be designed for running in a straight line, such as on a treadmill, or may include lateral features to improve grip when a user changes direction.

(57) In some embodiments, the plurality of bumps **160** may have different geometries across sole **112**. For example, a first bump **162** in toe region **120** may differ in height, slope, and/or size compared to a second bump **164** in heel region **124**. In some embodiments, the plurality of bumps **160** may differ in height, slope, and/or size within toe region **120**. In some embodiments, the plurality of bumps **160** may differ in height, slope, and/or size within midfoot region **122**. In some embodiments, the plurality of bumps **160** may differ in height, slope, and/or size within heel region **124**.

(58) In the illustrative embodiment shown in FIG. **8**, first bump **162** has a first height **161**. First height **161** may be measured from a first peak **163** at the apex of first bump **162** to a bottom **173** of an adjacent first valley **172** of the plurality of valleys **170**. Similarly, second bump **164** may have a second height **165** as shown, for example, in FIG. **9**. Second height **165** may be measured from a second peak **167** at the apex of second bump **164** to a bottom **175** of an adjacent second valley **174** of the plurality of valleys **170**.

(59) In some embodiments, second height **165** may be greater than first height **161**. In such embodiments, second bump **164** may provide a greater depth of cushioning in the heel region **124** of sole **112**. In such embodiments, first bump **162** may provide greater stiffness in the toe region **120** of sole **112**. In some embodiments, second height **165** may be greater than first height **161** by at least about 2 mm. In some embodiments, first height **161** may be greater than second height **165**. In some embodiments, first height **161** may be greater than second height **165** by at least about 2 mm. In some embodiment first and second bumps **162**, **164** may have equal height **161**, **165**.

(60) In some embodiments, first height **161** may be greater than or equal to about 0 mm to less than or equal to about 12 mm. In some embodiments, first height **161** may be greater than or equal to about 1 mm to less than or equal to about 12 mm. In some embodiments, first height **161** may be greater than or equal to about 4 mm to less than or equal to about 10 mm. In some embodiments, first height **161** may be greater than or equal to about 5 mm to less than or equal to about 7 mm.

(61) In some embodiments, second height **165** may be greater than or equal to about 5 mm to less than or equal to about 30 mm. In some embodiments, first height **161** may be greater than or equal to about 7 mm to less than or equal to about 15 mm. In some embodiments, first height **161** may be about 10 mm.

(62) In some embodiments, first bump **162** may comprise a flatter contour than at least one of the bumps of the plurality of bumps **160** located in the heel region of the sole **112** (for example, bump **164**). In some embodiments, first bump **162** may have shallower slope extending between bottom **173** of first valley **172** and first peak **163** of bump **162** compared to second bump **164** that may have steeper slope extending between bottom **175** of second valley **174** and second peak **167** of bump **164**.

(63) In some embodiments, first bump **162** may have a first slope angle **166** on the medial side of first bump **162** between about 5 degrees and about 30 degrees relative to plane **176** that is tangential to peak **163** of the first bump **162** and parallel to substantially flat upper surface **154**. In some embodiments, first bump **162** may have a second slope angle **168** on the lateral side of first bump **162** between about 5 degrees and about 30 degrees relative to plane **176**. In some embodiments, first slope angle **166** and second slope angle **168** may be different. In some embodiments, first slope angle **166** and second slope angle **168** may be equal.

(64) In some embodiments, second bump **164** may have third slope angle **169** on the medial side of second bump **164** between 50 degrees and 85 degrees relative to plane **177** that is tangential to peak

167 of the second bump **164** and parallel to substantially flat upper surface **154**. In some embodiments, second bump **164** may have fourth slope angle **171** on the lateral side of second bump **164** between 50 degrees and 85 degrees relative to plane **177**. In some embodiments, third slope angle **169** and fourth slope angle **171** may be different. In some embodiments, third slope angle **169** and fourth slope angle **171** may be equal. Examples of slope angles **166**, **168** of first bump **162** and slope angles **169**, **171** of second bump **164** are shown in FIGS. **8** and **9**.

(65) In some embodiments, first bump **162** may have a flatter contour with a continuous arc extending from adjacent valleys **172**. In some embodiments, second bump **164** may comprise a compound profile comprising steep side faces, an arc shaped ground-facing region, and radii connecting the steep side faces and arc shaped ground-facing region. As shown, for example, in FIGS. **8** and **9**, the connecting radii of second bump **164** may be smaller than an average radius of the continuous arc of the first bump **162**.

(66) In some embodiments first and second bumps **162**, **164** may have varying widths, from a lateral side to a medial side, that may be a function of bump height, slope angle, and/or radius. In some embodiments, a width of a bump on medial side **126** of sole **112** may be greater than a bump on lateral side **128** of sole **112**. In some embodiments, a width of a bump on lateral side **128** of sole **112** may be greater than a bump on medial side **126** of sole **112**. In some embodiments, the plurality of bumps **160** may be sized according to a gait pattern. In some embodiments, the plurality of bumps **160** may have greater width in a region of sole **112** where greater pressure is applied by a user according to the gait cycle or pattern. For example, the plurality of bumps **160** at medial side **126** of toe region **120** and lateral side **128** of heel region **124** may be wider than the plurality of bumps **160** at lateral side **128** of toe region **120** and medial side **126** of heel region **124**.

(67) In some embodiments, continuous skin **192** covering first bump **162** may have a greater surface area than continuous skin **192** covering second bump **164**. In some embodiments, continuous skin **192** covering first bump **162** may have a smaller surface area than continuous skin **192** covering second bump **164**.

(68) In some embodiments, lower side **152** may comprise a plurality of connecting ribs **180** that extend between continuous skins **192** of the plurality of bumps **160** as shown, for example in FIG. **7**. The plurality of connecting ribs **180** may be integrally formed (for example, integrally 3D printed) with continuous skins **192**. In some embodiments, the plurality of connecting ribs **180** may be integrally formed (for example, integrally 3D printed) on the plurality of interconnected unit cells **140** of mesh **130**. In embodiments comprising ribs **180**, the plurality of connecting ribs **180** may tie adjacent traction elements **190** together, via continuous skins **192**, such that the plurality of traction elements **190** may move together, or one traction element may resist movement of another traction element. Connecting ribs **180** may comprise flat strips of material that each extend from a first continuous skin **192** to a second continuous skin **192**. In some embodiments, connecting ribs **180** may extend on to medial side **126** and/or lateral side **128** and may be integrally formed with upper perimeter rim **153** on upper side **150**.

(69) In some embodiments, as shown for example in FIG. **7**, the plurality of connecting ribs **180** may comprise a wave or ripple shape between adjacent traction elements **190**. In some embodiments, the plurality of connecting ribs **180** may extend substantially straight between adjacent traction elements **190**. In some embodiments, the plurality of connecting ribs **180** may extend across the width of the sole **112**, from medial side **126** to lateral side **128**. In some embodiments, the plurality of connecting ribs **180** may extend across the length of the sole **112**, from toe region **120** to heel region **124**.

(70) In some embodiments, sole **112** may additionally comprise an outsole attached to each of continuous skin **192** of the plurality of traction elements **190**. In some embodiments, a spacer material may be secured to the continuous skins **192** and/or a forward toe portion **185** such that the outsole is attached to the spacer material rather than directly to the continuous skins **192** and/or a forward toe portion **185**. In such embodiments, sole **112** may be connected to the outsole via any of

various methods, such as by stitching, bonding, or by the use of adhesives, among other methods. The outsole may comprise a durable material, such as a natural or synthetic rubber, among others. (71) Another embodiment of sole **112** in accordance with the present disclosure is shown in FIGS. **10-12**. The sole **212** illustrated in FIGS. **10-12** is substantially similar to the sole **112** shown in FIGS. **1-9** and described herein. Accordingly, similar reference numbers in the **200** series indicate features that are common between sole **112** and sole **212**. The description of sole **112** is incorporated by reference to apply to sole **212**, except in instances when it conflicts with the specific description and the drawings of sole **212**. For example, sole **212** may comprise ledge **134** and/or upper perimeter rim **153**.

(72) In the illustrative embodiment shown in FIGS. **10-12**, sole **212** may comprise three-dimensional mesh **230** and a continuous ground-facing perimeter rim **232**. Mesh **230** may be formed in substantially the same way a mesh **130** and comprise upper side **250** and lower side **252**. Lower side **252** of mesh **230** may comprise a perimeter portion **260**, and continuous ground-facing perimeter rim **232** may be formed on perimeter portion **260**. In some embodiments, continuous ground-facing perimeter rim **232** may comprise a continuous skin surface **292** that covers the underlying mesh **230** at perimeter portion **260**.

(73) In some embodiments, continuous skin surface **292** may extend around at least two of the toe region **220**, the midfoot region **222**, and heel region **224** of lower side **252** of sole **212**. In some embodiments, continuous skin surface **292** may extend around each of the toe region **220**, the midfoot region **222**, and heel region **224** of lower side **252** of sole **212**.

(74) In some embodiments, unit cells **140** of mesh **230** may be exposed on lower side **252** in a middle portion **293** between the edges of continuous skin surface **292** of continuous ground-facing perimeter rim **232**.

(75) In some embodiments, as shown for example in FIGS. **10-12**, a recess **295** may be formed in middle portion **293** of lower side **252** of mesh **230**. In some embodiments, recess **295** may be formed in midfoot region **222** of middle portion **293**. Recess **295** extends into sole **212** relative to lower side **252** and continuous skin surface **292** of continuous ground-facing perimeter rim **232**. In some embodiments, recess **295** may be defined by a recessed region formed by unit cells **140** of mesh **230**. In some embodiments, recess **295** may be disposed in midfoot region **222** between medial and lateral sides of continuous skin surface **292** of continuous ground-facing perimeter rim **232**.

(76) In the illustrative embodiment shown in FIG. **12**, recess **295** has a depth **296**. Depth **296** may be measured from continuous skin surface **292** to a base **298** of recess **295**. In some embodiments, depth **296** may be greater than or equal to about 2 mm to less than or equal to about 15 mm. In some embodiments, depth **296** may be greater than or equal to about 5 mm to less than or equal to about 12 mm. In some embodiments, depth **296** may be about 10 mm. In some embodiments, recess **295** may comprise a maximum depth **296** that is greater than or equal to about 30% and less than or equal to about 70% of a total depth of sole **212** at the recess **295**.

(77) In some embodiments, continuous skin surface **292** may comprise a plurality of traction elements **294**. In such embodiments, traction elements **294** may serve to increase grip when sole **212** contacts the ground as shown during use. In some embodiments, as illustrated in for example FIGS. **10** and **11**, the plurality of traction elements **294** may comprise a ribbed traction pattern comprising a plurality of contoured ribs. In some embodiments, the plurality of traction elements **294** may comprise concentric ribbed circles, overlapping ribbed circles, wave shaped ribs, or any other suitable geometric shape. In some embodiments, the plurality of traction elements **294** may correspond to the sport the article of footwear **200** is used for. For example, the plurality of traction elements **294** may be designed for running in a straight line, such as on a treadmill, or may include lateral features to improve grip when a user changes direction.

(78) In some embodiments, lower side **252** may comprise a plurality of connecting ribs **280** that extend between medial and lateral sides of continuous skin surface **292** of continuous ground-

facing perimeter rim **232** as shown, for example in FIGS. **10** and **11**. In some embodiments, the plurality of connecting ribs **280** may be integrally formed (for example, integrally 3D printed) with mesh **230**. In such embodiments, the plurality of connecting ribs **280** may be integrally formed on the plurality of interconnected unit cells **140** of mesh **230**. In some embodiments, the plurality of connecting ribs **280** may be integrally formed (for example, integrally 3D printed) with continuous skin surface **292**.

(79) In some embodiments, the plurality of connecting ribs **280** may extend across middle portion **293** of lower side **252** and tie opposite medial and lateral sides of continuous skin surface **292** together. In such embodiments, the plurality of connection ribs **280** may resist deformation of one side of the perimeter rim **232** compared to the other side. In some embodiments, as shown for example in FIGS. **10** and **11**, the plurality of connecting ribs **280** comprise wave or ripple shape between adjacent sides of the continuous skin surface **292**. In some embodiments, the plurality of connecting ribs **280** may extend substantially straight between adjacent sides of continuous skin surface **292**. In some embodiments, the plurality of connecting ribs **280** may extend across the width of the sole **212**, from medial to lateral sides. Connecting ribs **280** may comprise flat strips of material that each extend from a medial side to a lateral side of continuous skin surface **292**.

(80) Where a range of numerical values comprising upper and lower values is recited herein, unless otherwise stated in specific circumstances, the range is intended to include the endpoints thereof, and all integers and fractions within the range. It is not intended that the disclosure or claims be limited to the specific values recited when defining a range. Further, when an amount, concentration, or other value or parameter is given as a range, one or more ranges, or as list of upper values and lower values, this is to be understood as specifically disclosing all ranges formed from any pair of any upper range limit or value and any lower range limit or value, regardless of whether such pairs are separately disclosed. Finally, when the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range recites “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.”

(81) As used herein, the term “about” refers to a value that is within +10% of the value stated. For example, about 10% can include any percentage between 9% and 11%.

(82) It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present invention(s) as contemplated by the inventors, and thus, are not intended to limit the present invention(s) and the appended claims in any way.

(83) The foregoing description of the specific embodiments will so fully reveal the general nature of the invention(s) that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, and without departing from the general concept of the present invention(s). Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance herein.

(84) The breadth and scope of the present invention(s) should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents

Claims

1. An article of footwear, comprising: a sole comprising: a three-dimensional mesh extending from a toe region to a heel region of the sole, the mesh comprising: a lattice structure comprising a plurality of interconnected unit cells, each interconnected unit cell comprising a plurality of struts defining a three-dimensional structure and a plurality of nodes at which one or more unit cell struts are connected; a ground-facing portion comprising a plurality of traction elements and located at a lower side of the mesh opposite an upper side; and a ledge formed on the upper side of the mesh and comprising a continuous side surface extending toward the lower side and along at least one of an outer lateral side or an outer medial side of the mesh, wherein the continuous side surface faces outwardly away from the outer lateral side or the outer medial side of the mesh; an upper coupled to the mesh; and a joining element extending between and coupling the outwardly facing continuous side surface of the ledge to the upper.
2. The article of footwear of claim 1, wherein the mesh, the ledge, and the ground-facing portion are integrally formed as a single piece.
3. The article of footwear of claim 1, wherein the continuous side surface of the ledge extends around the heel region of the sole from the outer lateral side to the outer medial side of the mesh.
4. The article of footwear of claim 3, wherein the ledge further comprises a solid top surface formed on an upper surface of the upper side of the mesh, wherein the solid top surface is disposed above the continuous side surface.
5. The article of footwear of claim 1, wherein the joining element is coupled to the continuous side surface of the ledge by adhesive.
6. The article of footwear of claim 1, wherein the lower side of the mesh comprises a plurality of bumps and a plurality of valleys positioned between respective bumps, and wherein the plurality of traction elements of the ground-facing portion are formed on each of the plurality of bumps.
7. The article of footwear of claim 6, wherein each of the plurality of traction elements comprises a separate continuous skin that covers a portion a respective one of the plurality of bumps.
8. The article of footwear of claim 1, wherein the lower side of the mesh comprises a ground-facing perimeter rim and a middle portion within the ground-facing perimeter rim, and wherein the plurality of traction elements of the ground-facing portion are formed on the ground-facing perimeter rim.
9. The article of footwear of claim 8, wherein the middle portion is recessed relative to the perimeter rim.
10. The article of footwear of claim 8, wherein the ground-facing perimeter rim comprises a continuous surface skin that extends around each of the toe region, a midfoot region, and the heel region of the sole and the plurality of traction elements are formed on the continuous surface skin.
11. A sole for an article of footwear, the sole comprising: a three-dimensional mesh extending from a toe region to a heel region of the sole, the mesh comprising: a lattice structure comprising a plurality of interconnected unit cells, each interconnected unit cell comprising a plurality of struts defining a three-dimensional structure and a plurality of nodes at which one or more unit cell struts are connected, and an undulating lower side comprising a plurality of bumps and a plurality of valleys positioned between respective bumps, wherein the bumps are at least partially defined by one or more unit cells of the lattice structure; and a plurality of traction elements formed on the plurality of bumps of the undulating lower side of the mesh such that the plurality of traction elements are ground-facing, wherein a first bump of the plurality of bumps located in the toe region comprises a first height measured between a first peak of the first bump and a first bottom of an adjacent valley of the plurality of valleys, a second bump of the plurality of bumps located in the heel region comprises a second height measured between a second peak of the second bump and a second bottom of an adjacent valley of the plurality of valleys, and wherein the second height is different from the first height.
12. The sole of claim 11, wherein the second height is larger than the first height.

13. The sole of claim 11, wherein each one of the plurality of traction elements comprises a separate continuous skin that covers a portion of a respective one of the plurality of bumps.
14. The sole of claim 13, wherein the continuous skin of each traction element comprises a ribbed traction pattern.
15. The sole of claim 11, wherein at least one of the bumps of the plurality of bumps located in the toe region of the sole comprises a flatter contour than at least one of the bumps of the plurality of bumps located in the heel region of the sole.
16. The sole of claim 11, wherein the sole further comprises a plurality of connecting ribs formed on the undulating lower side of the mesh and extending between adjacent traction elements of the plurality of traction elements.
17. A sole for an article of footwear, the sole comprising: a three-dimensional mesh extending from a toe region to a heel region of the sole, the mesh comprising: a lattice structure comprising a plurality of interconnected unit cells, each interconnected unit cell comprising a plurality of struts defining a three-dimensional structure and a plurality of nodes at which one or more unit cell struts are connected; and a continuous ground-facing rim formed on a perimeter portion of a lower side of the mesh, wherein the continuous ground-facing rim extends around each of the toe region, a midfoot region, and the heel region of the sole, wherein the continuous ground-facing rim comprises a plurality of traction elements formed on a ground-facing surface of the rim, wherein the unit cells of the mesh form a recess in the midfoot region of a middle portion of the mesh between the ground-facing rim, and wherein a first plurality of connecting ribs are formed on the ground-facing side of the recess and extend across the recess to connect a lateral side of the ground-facing rim to a medial side of the ground-facing rim.
18. The article of footwear of claim 17, wherein the sole further comprises a second plurality of connecting ribs formed on the mesh and extending through a middle portion of the lower side of the toe region or the heel region to connect the lateral side of the ground-facing rim to the medial side of the ground-facing rim.
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